

Machine Learning

11. Unsupervised Learning 2

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Today, we will learn...

- Unsupervised Learning 2
 - What is 'dimensionality reduction'?
 - Principal Component Analysis (PCA)



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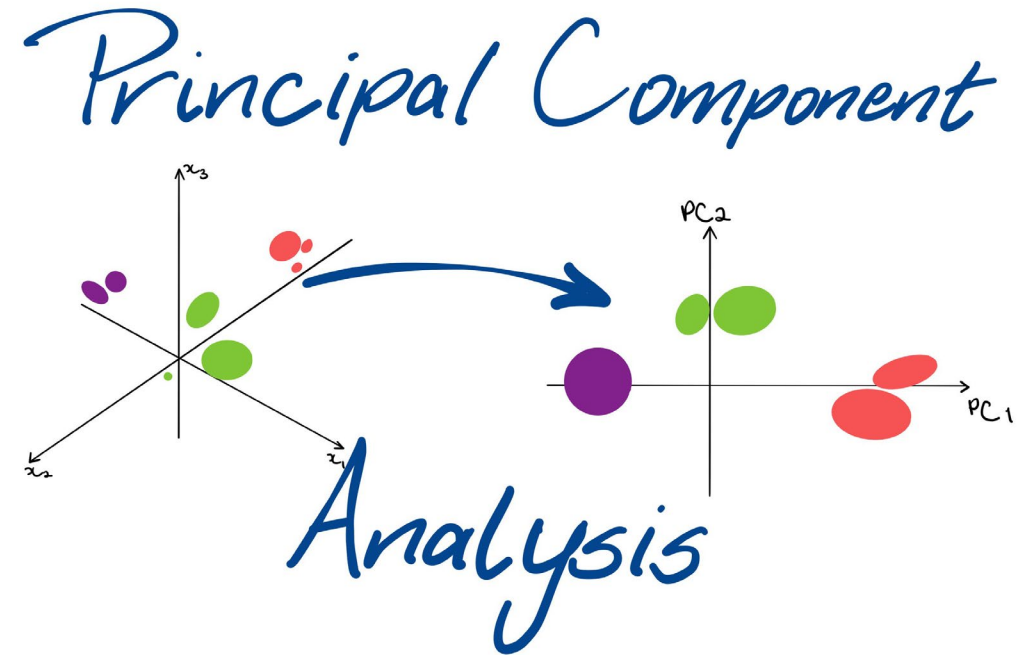
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- **Motivation**
- Principal Component Analysis



Big & High Dimensional Data (1/2)

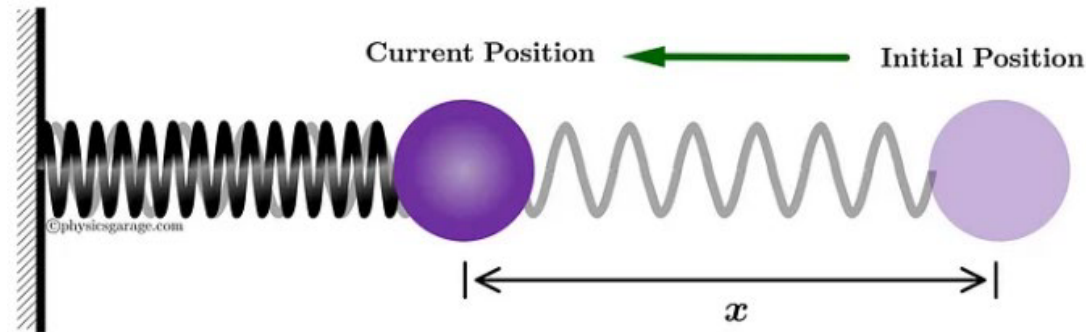
- High-dimensions == A lot of data features
- E.g., document classification
 - Features per document may include thousands of words/unigrams as well as contextual information.
- Surveys – **Netflix** (2016 October)
 - 480,189 users x 17,770 movies

	Movie 1	Movie 2	Movie 3	Movie 4	Movie 5
Burt	4	?	?	1	4
Lisa	?	5	3	3	?
Homer	2	6	4	?	6

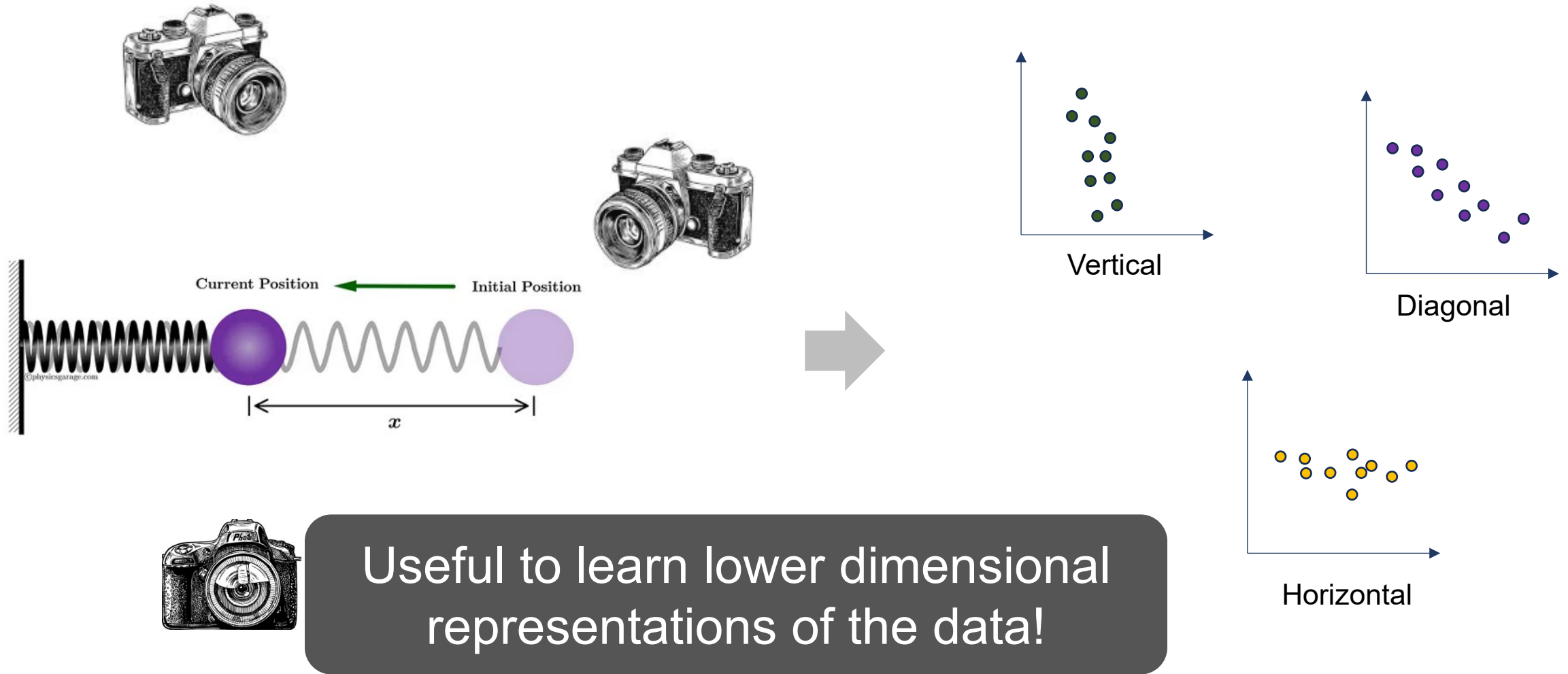
Motivation: Toy Example (1/2)

- Sometimes, too much information is not helpful!
- Say a ball is attached to an oscillating spring.
 - Could you simply understand how the ball will move?

공의 움직임에 대한 관측(데이터)가 있어야 기계학습으로 모델링 가능

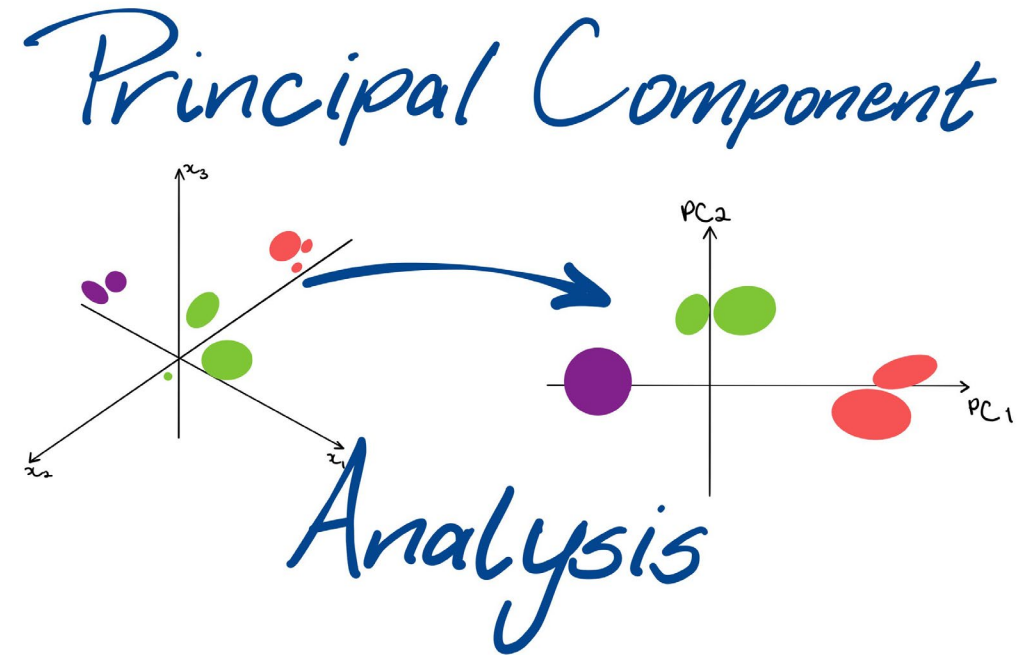


Motivation: Toy Example (2/2)



- Motivation
- **Principal Component Analysis**

Non-Parametric, 모델이 없음

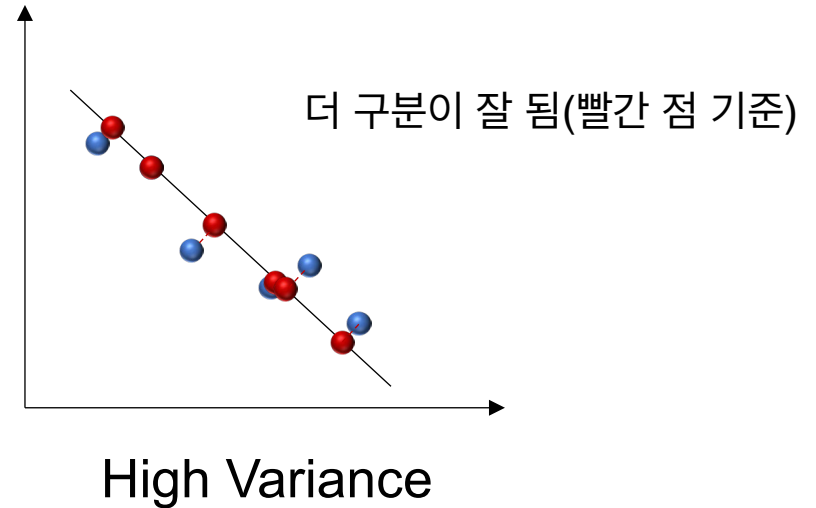
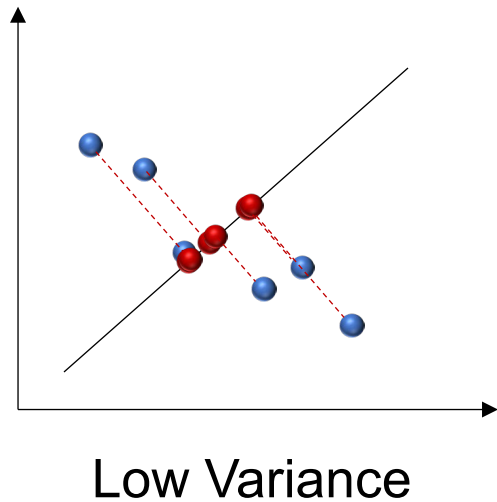


Learning Representations

- ***Principal Component Analysis*** (PCA): A powerful unsupervised learning techniques for extracting (potentially lower dimensional) structure from high dimen datasets.
- Useful for:
 - Data visualization
 - More efficient use of resources (e.g., time, memory, or communication)
 - Statistically, better generalization
 - Noise removal

Principal Components?

- We will 'project' our data on a lower-dimensional space.
- The natural question is where to project the data.



We want to find vectors that maximize data variance!

Background: Covariance Matrix

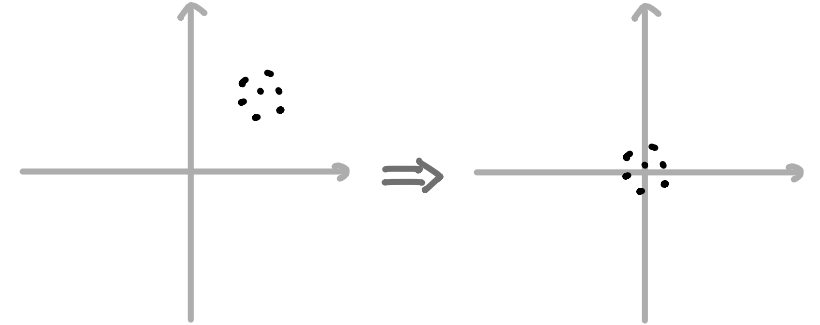
- Covariance Matrix

$$A = \{a_1, a_2, \dots, a_n\}, B = \{b_1, b_2, \dots, b_n\}$$

$$\sigma_A^2 = \frac{1}{n} (\sum_i a_i^2 - (\sum_i a_i)^2), \sigma_B^2 = \frac{1}{n} (\sum_i b_i^2 - (\sum_i b_i)^2)$$

$$\sigma_A^2 = \frac{1}{n} \sum_i a_i^2, \sigma_B^2 = \frac{1}{n} \sum_i b_i^2, \text{ if each sample is normalized to have zero mean.}$$

$$\sigma_{AB}^2 = \frac{1}{n} \sum_i (a_i - \mu_A)(b_i - \mu_B) = \frac{1}{n} \sum_i a_i b_i$$



- Multi-dimensional Covariance Matrix

$$\mathbf{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_m \end{bmatrix}, \text{ then } \mathbf{C}_X = \frac{1}{n} \mathbf{X} \mathbf{X}^T$$

고차원 데이터 X에 대해 표준화 시에만 만족

Assumptions

1. Linearity

- We will assume the linearity of data so that the problem is defined as a change of basis.

2. Large variances have important structure.

- We will assume a high Signal-to-Noise Ratio is preferable.

3. The principal components are orthogonal.

- We will find the minimal set of independent ways to represent the given data.

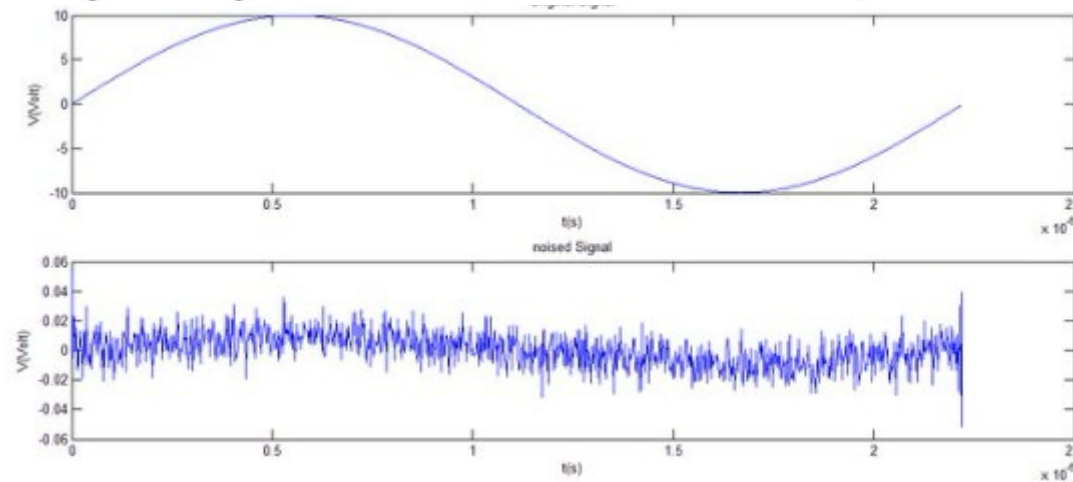
Assumption 1: Linearity

- Why do we need linearity? 기본 트랜스포즈 알고리즘은 시간복잡도 매우 높음, 따라서 선형 변환 사용한 트랜스포즈 필요
 - Our goal is to project the data onto a lower-dimensional subspace that captures the maximum variance. 모든 차원이 선형이어야만 선형 변환으로 트랜스포즈 가능
 - This process relies on linear transformations, assuming the underlying data structure is linear.
 - In addition, the transformation can be efficiently implemented using matrix operations only when the data is linearly structured.

Assumption 2: Large variances have important structure.

- From statistics' perspective,
 - Signal: The wider the data are distributed, the more the meaningful patterns or changes.
 - Noise: The small variance indicates that the data does not change much, and it can be considered as ill-observed noise.

We will assume a high Signal-to-Noise Ratio is preferable.



Assumption 3: Orthogonality

- Redundancy Removal (Minimal Representation)
 - Our goal is to find the tightest minimal representation of data.
- The optimal solution of $\operatorname{argmax}_{x'} (Var(x))$
 - All individual eigenmode should maximize variance of different dimensions to maximize the total variance.

